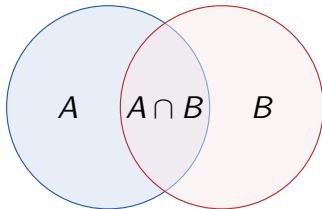


Inclusion and Exclusion

Section 2.5

Harris–Hirst–Mossinghoff, *Combinatorics and Graph Theory*



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Today's lecture

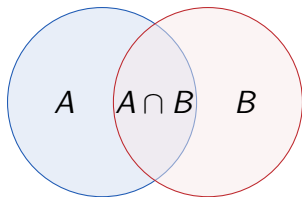
1. Counting Unions

2. The General Formula

3. Derangements and Surjections

4. Further Applications

Counting overlapping sets



$$|A \cup B| = |A| + |B| - |A \cap B|.$$

Three sets

Formula

$$\begin{aligned} |A \cup B \cup C| = & |A| + |B| + |C| \\ & - |A \cap B| - |A \cap C| - |B \cap C| \\ & + |A \cap B \cap C|. \end{aligned}$$

Elements belonging to all three sets are added three times, subtracted three times, and finally added once.

Inclusion and exclusion

Principle of inclusion and exclusion

For finite sets $A_1, \dots, A_r$,

$$\left| \bigcup_{i=1}^r A_i \right| = \sum_{\emptyset \neq I \subseteq [r]} (-1)^{|I|+1} \left| \bigcap_{i \in I} A_i \right|.$$

Add single sets, subtract pairwise intersections, add triple intersections, and continue with alternating signs.

Proof by following one element

Proof

An element lying in exactly $t \geq 1$ of the sets is counted

$$\binom{t}{1} - \binom{t}{2} + \cdots + (-1)^{t+1} \binom{t}{t}$$

times. Since $\sum_{j=0}^t (-1)^j \binom{t}{j} = 0$, this alternating sum is 1.

An element in none of the sets is never counted. Thus the formula counts the union exactly once.

Counting objects avoiding every bad property

Complement form

If all $A_i \subseteq U$, then

$$\left| U \setminus \bigcup_{i=1}^r A_i \right| = \sum_{I \subseteq [r]} (-1)^{|I|} \left| \bigcap_{i \in I} A_i \right|,$$

where the intersection for $I = \emptyset$ means U .

Let A_i describe a forbidden property.

Integers divisible by 2, 3, or 5

Question

How many integers from 1 to 100 are divisible by at least one of 2, 3, 5?

The individual counts are 50, 33, 20. Pairwise intersections count multiples of 6, 10, 15, and the triple intersection counts multiples of 30.

The divisibility calculation

$$50 + 33 + 20 - 16 - 10 - 6 + 3 = 74.$$

Hence $100 - 74 = 26$ integers in the range are divisible by none of 2, 3, 5.

Intersections are counted using least common multiples; multiplying divisors works here because the original divisors are pairwise coprime.

Derangements

Derangement

A derangement is a permutation with no fixed point. Let D_n be the number of derangements of $[n]$.

Example

For $n = 3$, the derangements in one-line notation are 231 and 312. Thus $D_3 = 2$.

Derangements: identify the bad sets

Let U be the set of all $n!$ permutations, and let

$$A_i = \{\pi : \pi(i) = i\}.$$

If a specified set of k positions is fixed, the remaining positions can be permuted in $(n - k)!$ ways.

There are $\binom{n}{k}$ such sets of positions.

The derangement formula

Theorem

$$D_n = \sum_{k=0}^n (-1)^k \binom{n}{k} (n-k)! = n! \sum_{k=0}^n \frac{(-1)^k}{k!}.$$

Proof

Apply inclusion and exclusion to the sets of permutations fixing each position.

Small derangement numbers

n	0	1	2	3	4	5	6
D_n	1	0	1	2	9	44	265

The empty permutation gives $D_0 = 1$.

For example, $D_4 = 24 - 24 + 12 - 4 + 1 = 9$.

A derangement recurrence

Theorem

For $n \geq 2$,

$$D_n = (n - 1)(D_{n-1} + D_{n-2}).$$

Look at the cycle containing n . Its length is either 2 or at least 3.

Proof of the derangement recurrence

Proof

If n lies in a 2-cycle, choose its partner in $n - 1$ ways and derange the other $n - 2$ elements.

If its cycle has length at least 3, remove n from that cycle. This leaves a derangement on $n - 1$ elements. Conversely, insert n immediately after any of the $n - 1$ elements in a cycle of such a derangement.

The two cases give $(n - 1)D_{n-2} + (n - 1)D_{n-1}$.

Exactly j fixed points

Formula

The number of permutations of $[n]$ with exactly j fixed points is

$$\binom{n}{j} D_{n-j}.$$

Proof

Choose the fixed points, then derange everything else.

Using $(n - j)!$ instead would allow additional fixed points.

The probability of no fixed points

$$\frac{D_n}{n!} = 1 - 1 + \frac{1}{2!} - \frac{1}{3!} + \cdots + \frac{(-1)^n}{n!}.$$

As $n \rightarrow \infty$, this converges to e^{-1} .

The alternating-series bound gives

$$\left| D_n - \frac{n!}{e} \right| \leq \frac{1}{n+1}.$$

Question

How many functions from an n -element set onto a k -element set are there?

There are k^n functions in all. Let A_i be the functions omitting target value i .

If j specified target values are omitted, there are $(k - j)^n$ functions.

Counting onto functions

Formula

For $n, k \geq 1$, the number of surjections is

$$\sum_{j=0}^k (-1)^j \binom{k}{j} (k-j)^n.$$

Example

Onto functions from a 4-set to a 3-set number

$$3^4 - 3 \cdot 2^4 + 3 \cdot 1^4 = 36.$$

Labeled boxes with none empty

Assigning n distinct objects to k labeled nonempty boxes is the same as choosing a surjection.

Later, dividing the number of surjections by $k!$ will count partitions of an n -element set into k nonempty unlabeled blocks.

Bounded integer solutions

Question

How many nonnegative integer solutions satisfy

$$x_1 + \cdots + x_r = N, \quad x_i \leq b?$$

Without the upper bounds there are $\binom{N+r-1}{r-1}$ solutions.

Let the bad property A_i be $x_i \geq b + 1$.

Intersections of the bad properties

Proof

If a specified set of j coordinates violates the bound, subtract $b + 1$ from each of those coordinates. This gives nonnegative solutions with total $N - j(b + 1)$.

The number is

$$\binom{N - j(b + 1) + r - 1}{r - 1}$$

when the remaining total is nonnegative, and zero otherwise.

The bounded-solution formula

Formula

The number is

$$\sum_{j=0}^{\min(r, \lfloor N/(b+1) \rfloor)} (-1)^j \binom{r}{j} \binom{N - j(b+1) + r - 1}{r - 1}.$$

We will obtain the same alternating sum from generating functions.

A small bounded example

Question

Count solutions of $x_1 + x_2 + x_3 = 8$ with $0 \leq x_i \leq 3$.

$$\binom{10}{2} - 3\binom{6}{2} + 3\binom{2}{2} = 45 - 45 + 3 = 3.$$

They are $(2, 3, 3), (3, 2, 3), (3, 3, 2)$.

Chromatic polynomials by inclusion and exclusion

For a graph $G = (V, E)$, start with all $k^{|V|}$ assignments of colors. For each edge $e = uv$, let A_e be the assignments with u, v the same color.

If all edges in $F \subseteq E$ impose equality, every component of (V, F) is monochromatic. There are $k^{c(F)}$ assignments, where isolated vertices count as components.

$$P_G(k) = \sum_{F \subseteq E} (-1)^{|F|} k^{c(F)}.$$

The triangle again

For K_3 , edge subsets of sizes 0, 1, 2, 3 contribute

$$P_{K_3}(k) = k^3 - 3k^2 + 3k - k = k(k-1)(k-2).$$

Inclusion and exclusion recovers the proper-coloring count by correcting monochromatic edges.

Question

How many words of length 5 over $\{a, b, c\}$ use all three letters?

Count words missing at least one letter, then subtract from the total.

$$3^5 - \binom{3}{1}2^5 + \binom{3}{2}1^5 = 243 - 96 + 3 = 150.$$

A word missing a specified letter has 2^5 possibilities. A word missing two specified letters has only one possibility.

- Alternating intersection counts correct overlaps exactly.
- Describe forbidden properties before applying the formula.
- Derangements, surjections, and bounded allocations are standard applications.
- The same principle also produces chromatic polynomials.

Generating Functions